Every transformer block has two steps, attention is first then feed-forward. This stage is when the word thinks about the information it has received.

What it does

- A word's vector is checked against a big list of learned patterns. ("is this a past tense? An animal?")
- Then it rewrites/adjust the word based on which patterns are found.
- Where a high number = strongly present, 0 ≈ absent.

How it works:

- The word vector is expanded from 512 → 2048, via matrix multiplication. This is done to allow more space to check for patterns.
- Then the ReLU (Rectified Linear Unit) keeps positive numbers and turn negative ones to 0.
- Finally a second matrix is multiplied to shrink the vector back to 512 dimensions.